ANISH KUMAR KHARWAR

Senior Software Engineer | Distributed Systems · Data Pipelines · Scalable Microservices · Cloud (GCP/Azure)

E-mail: anishk7475@gmail.com | Phone: +91 7905395419
LinkedIn: anishk74 | GitHub: anishk74 | Portfolio: https://anishk74.github.io

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Experience

Senior Software Engineer | Nference
Python, Bash, Google Cloud Platform (GCP), Microsoft Azure, Redis, Django

August 2021 – September 2025

- Designed and deployed **4** new microservices within a publisher-subscriber system to process and transform **480TB+** of high-volume and multi-format data, ensuring scalable and reliable ingestion across diverse datasets.
- Extended image parsing microservices to support **700GB+** of PDF datasets by implementing logic to handle overlapping characters and unordered text segments, enabling accurate text extraction from the files.
- Enhanced structured data microservices to support **80+** additional data categories by analyzing raw datasets and extending processing workflows, enabling broader data coverage and improved system capabilities.
- Developed a pipeline to aggregate and analyze data loss during processing across **45TB+** structured datasets, optimizing processing logic and reducing data loss from **23%** to **9%**.
- Integrated and optimized a new image model into image processing microservices, enabling processing of **12M+** files.
- Collaborated with client engineering teams to troubleshoot deployment issues, resolve incorrect data formats and address processing errors in deployed services, ensuring smooth onboarding for multiple deployments.
- Developed and maintained **60+** backend APIs for authentication and role-based access control, implementing secure management of users, groups, organizations, and applications to enforce granular permission models.

Computer Vision and Machine Learning Engineer | Awiros
Python, C++, MXNet

June 2021 – July 2021

- Designed data preprocessing pipelines for large-scale image datasets (**494K+** and **43K+** files), implementing logic for distribution analysis, aspect ratio handling, and quality checks to ensure robust downstream processing.
- Implemented backend automation for the ML lifecycle, including model creation, training, deployment, and inference, integrating with the UI to deliver streamlined, production-ready workflows.

Backend Intern | Vaultedge Software
Python, Flask, MongoDB, Grafana

March 2021 – May 2021

- Optimized job scheduling logic by migrating from FIFO to a priority queue system, improving task throughput and reducing latency for high-priority operations.
- Designed and implemented **6** REST APIs in Flask to serve performance data from MongoDB to Grafana dashboards, enabling low-latency and real-time visualization.
- Created **20+** system performance metrics and equations for monitoring and display, improving operational visibility.

Skills

Languages: Python, Bash
Backend: Django, Flask, REST APIs
Architecture: Distributed Systems, Microservices, Pub/Sub
Cloud & DevOps: GCP, Azure, Linux, Docker, CI/CD, Git
Databases & Queues: Redis, MySQL, MongoDB, RabbitMQ
Observability: Grafana, Prometheus

Education

2017 – 2021
Bachelor of Technology, Computer Science and Engineering
Indian Institute of Information Technology Kalyani
Cumulative GPA: 8.29

Interests

Distributed Computing, Cloud-Native Scalability, High-Throughput & Low-Latency Systems